① Define a vector space over $\mathbb{R}$. State the two closure properties required for a set to qualify as a vector space?

Ans:-

A vector space over $\mathbb{R}$ is a set $V$ together with two operations, ~~vect~~ vector addition and scalar multiplication (where scalars are real numbers $\mathbb{R}$), that satisfy the vector space axioms.

→ Two closure properties:-

① closure under addition: If $u, v \in V$, then $u+v \in V$

② closure under scalar multiplication:

if $u \in V$ and $c \in \mathbb{R}$, then $cu \in V$

② What is a vector in $R^n$? Explain it's structure and how it's represents features in machine learning?

→ A vector in $R^n$ is an ordered list of n real Numbers:

$$x = (x_1, x_2, x_3, ---, x_n)$$

where each $x_i \in R$.

→ structure:-

① It has n components

② Each component represents a coordinate in n dimensional space.

③ It can be written as a row vector:- $(x_1, x_2, x_3, ---, x_n)$

or as a column vector:

$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

③ Define a subspace of a vector space. State the three conditions required to prove that a subset is a subspace.

Ans:- A subspace of a vector space V is a subset $W \subseteq V$ that is itself a vector space under the same operations of addition and scalar multiplication defined on V.

→ Three conditions to prove a subspace:

a subset W is a subspace of V, we must verify: ① contains the zero vector:

$$0 \in W$$

② closed under addition:-

if $u, v \in W$, then, $u+v \in W$

③ closed under scalar multiplication:-

if $u \in W$ and $c \in R$, then $cu \in W$

If these three conditions are satisfied, then W is a subspace of V.

④ What is a linear combination of vectors? write its general mathematical form?

Ans:- A linear combination of vectors is an expression formed by multiplying each vector by a scalar and then adding the results. If $U_1, U_2, \ldots, U_K$ are vectors and $c_1, c_2, \ldots, c_K$ are scalars, then their linear combination is:

$$c_1 v_1 + c_2 v_2 + \cdots + c_K v_K = \sum_{i=1}^{K} c_i v_i$$

where, $v_i$ are vectors

$c_i \in R$ are scalars.

⑤ Define the span of a set of vectors. What does span represent geometrically?

→ The span of a set of vectors is the set of all possible linear combinations of those vectors. If $u_1, u_2, \ldots, u_k$ are vectors, then:

$$\text{span}\{u_1, u_2, \ldots u_k\} = \{c_1 v_1 + c_2 u_2 + \ldots + c_k u_k \mid c_1, c_2, \ldots c_k \in \mathbb{R}\}$$

Geometric meaning:-

ⓐ one non-zero vector → a line through the origin

ⓑ Two linear independent vectors → a plane through the origin.

ⓒ Three independent vectors in $\mathbb{R}^3$ → The entire 3D space.

⑤ what is linear independence? write the mathematical condition that determines whether a set of vectors is linearly independent?

Ans:-

Linear independence means that one of the vectors in a set can be written as a linear combination of the others.

A set of vector $u_1, u_2, \ldots, u_K$ is linearly independent if the only solution to the equation $c_1u_1 + c_2u_2 + \cdots + c_Ku_K = 0$

$$c_1 = c_2 = \cdots = c_K = 0$$

(where $0$ is the zero vector)

⑦ Explain linear dependence and describe why it creates problems in machine learning?

Ans:-

Linear dependence mean that in a set of vector, at least one vector can be written as a linear combination of the others.

mathematically, the vectors $U_1, U_2, \ldots, U_K$ are linearly dependent if there exist scalar $C_1, C_2, \ldots, C_K$, not all zero, such that

$$c_1 U_1 + c_2 U_2 + \cdots + c_K U_K = 0$$

In (ml) vectors usually represent features. Linear dependence means some feature are redundant.

(i) multicollinearity (ii) Unstable coefficients

(iii) Matrix inverse proble

(iv) overfiting Risk.

⑧ Define a basis of a vector space? What two properties must a set of vectors satisfy to form a basis?

Ans:- Two Properties:-

ⓐ Linearly independent:- The vectors must be linearly independent.

ⓑ Span the space:- The vectors must span the vector space V, meaning every vector in V can be written as a linear combination of them.

(5) What is ment by the dimension of a vector space? How is it related to basis?

Ans:- The dimension of a vector space is the number of vectors in it's basis.

Relation to Basis:-

* A basis is a set of linearly independent vectors that span the space.

* The dimension tell us how many such independent directions are needed to describe the space.

Dimesion of V = Number of vectors in any basis of V

(10) What does matrix rank indicates about a set of vectors? How is it related to linear independence?

Ans:- The Rank of a matrix is the maximum number of linearly independent rows or column in that matrix.

* if vectors are written as columns of a matrix:-

(a) The Rank tells us how many of those vectors are linearly independent.

(b) It shows the dimension of the span of those vectors.